

## Step 1: Video Viewing and Vocabulary

### 1. Watch the Video

(<https://www.youtube.com/watch?v=G5t7T6krW9U>).

Focus on any direct statements, commands, or questions mentioned.

### 2. Key Vocabulary

Extract the following technical and formal terms from the video:

- Query
- Instruction
- Permission
- Protocol
- Deployment
- Feedback
- Implementation

### 3. Task: Write a definition for each word based on its context in the video.

## Step 2: Questions on Reported Speech

### 1. Direct Questions to Reported Questions Convert the following into reported speech:

- "What is the best way to implement this system?"
- "Can you explain the process again?"
- "Did you find any errors during testing?"

### 2. Example Answer:

*They asked what the best way to implement the system was.*

## Understanding Cybersecurity

In our increasingly digital world, cybersecurity plays a vital role in protecting systems, networks, and data from potential cyber threats. For you as students, having a strong grasp of cybersecurity concepts is essential, as you will likely manage critical infrastructures and sensitive information in their careers.

One fundamental concept in cybersecurity is vulnerability. A vulnerability is a weakness in a system, software, or hardware that can be exploited by hackers. For example, failing to update an operating system or using weak passwords creates opportunities for cyberattacks. Students should learn to mitigate these risks by applying patches and enforcing strong authentication protocols, such as multi-factor authentication (MFA).

Another critical aspect is encryption, which involves converting readable data into a secure format to prevent unauthorized access. This ensures that even if data is intercepted, it cannot be understood. Using tools like firewalls, antivirus software, and intrusion detection systems (IDS) helps create robust layers of defense against malware, phishing, and other attacks.

Social engineering is another significant threat. Attackers often manipulate individuals to gain access to confidential information. Recognizing phishing emails, suspicious links, and unusual requests is crucial for preventing data breaches caused by human error.

In professional environments, network monitoring is essential for identifying potential threats in real time. Additionally, maintaining regular backups ensures that organizations can recover their data in case of a security breach or ransomware attack. Students should also familiarize themselves with incident response plans, which outline steps to take when a security issue arises.

By mastering these concepts, ASIX and DAM students can develop the skills needed to protect both personal and organizational systems. Cybersecurity is not just a technical skill but an ongoing commitment to learning and staying ahead of emerging threats. Stay vigilant, stay secure!

## QUESTIONS ON THE TEXT

1. What is the primary role of cybersecurity in today's digital world?
2. What is a vulnerability, and why is it important to address it in cybersecurity?
3. How can weak passwords and outdated operating systems be exploited by hackers?
4. What is encryption, and how does it protect sensitive information?
5. Which tools are mentioned as essential for creating a strong defense against cyberattacks?
6. What is social engineering, and how do attackers use it to access confidential information?
7. Why is it important to recognize phishing emails and suspicious links?
8. What are intrusion detection systems (IDS), and how do they help in professional environments?
9. How can backups help organizations recover from a security breach or ransomware attack?
10. Why is lifelong learning important in the field of cybersecurity?

## Part 2: Analyze a Real-Life Cybersecurity Breach

### Objective:

You will research a real-life cybersecurity breach, summarize its causes, consequences, and the measures taken to prevent future incidents.

### Instructions:

1. **Research:** Choose a significant cybersecurity breach (e.g., Target, Equifax, or SolarWinds attacks).

*The Target and Equifax data breaches were two of the most significant cybersecurity incidents in recent history, with far-reaching consequences for millions of consumers and the companies involved.*

Look for information about:

- How the breach occurred (e.g., vulnerabilities exploited, social engineering, etc.).
  - The impact of the breach on the company and its customers.
  - The measures implemented to strengthen cybersecurity afterward.
2. **Write a Report:** Prepare a 200-300 word summary including:
    - A brief introduction to the breach.
    - Key technical details about how the breach happened.
    - The main lessons learned and best practices to prevent such breaches in the future.
  3. **Present Your Findings:** Prepare to share your findings in class and submit them via Moodle.

## LISTENING TASK

The video titled "Top 10 Cyber Security Tools 2024"

<https://www.youtube.com/watch?v=r1tTCmRNBPo> provides an overview of essential cybersecurity tools that are crucial for protecting digital assets.

### Comprehension Questions:

1. What is the primary purpose of the cybersecurity tools mentioned in the video?
2. Can you name three cybersecurity tools highlighted in the video and briefly describe their functions?
3. How does a firewall contribute to network security, according to the video?
4. What role does antivirus software play in protecting computer systems?
5. Why is it important to use intrusion detection systems?

## Relevant Vocabulary

1. **Firewall:** A system designed to prevent unauthorized access to or from a private network.
2. **Antivirus Software:** Programs that detect, prevent, and remove malware from systems.
3. **Intrusion Detection System (IDS):** A tool for monitoring network traffic to identify suspicious activity.
4. **Penetration Testing:** A process where cybersecurity professionals simulate attacks to find vulnerabilities in a system.
5. **Encryption:** The process of converting data into a secure format to protect it from unauthorized access.
6. **Authentication:** Verifying the identity of a user or device.
7. **Multi-Factor Authentication (MFA):** A security system that requires multiple forms of identification to access.
8. **Phishing:** Fraudulent attempts to steal sensitive information, often through deceptive emails or websites.
9. **Malware:** Malicious software designed to damage or disrupt systems, including viruses, worms, and ransomware.
10. **Endpoint Security:** Measures taken to protect devices such as laptops, smartphones, and tablets from cyber threats.
11. **Vulnerability:** A weakness in software, hardware, or processes that can be exploited.
12. **Patch Management:** The process of updating software to fix vulnerabilities and improve security.
13. **Ransomware:** A type of malware that locks systems or data until a ransom is paid.
14. **Data Breach:** An incident where sensitive, protected, or confidential information is accessed or disclosed without authorization.
15. **Cyber Threat:** Any potential malicious attack that seeks to damage data, steal information, or disrupt digital operations.

## Additional Learning Activity

You are encouraged to **expand this list** by researching additional relevant cybersecurity terms. For example:

- **Zero-Day Vulnerability:** A flaw in software that is unknown to the vendor and exploited by attackers.
- **Incident Response:** A structured approach for managing and responding to cybersecurity incidents.
- **DDoS (Distributed Denial of Service):** A type of cyberattack that overwhelms a network with excessive traffic to make it unavailable.

### **Assessment**

**1. Comprehension Questions (30%)**

- Evaluate understanding of cybersecurity concepts from the article and video.
- Focus on accuracy, detail, and relevant vocabulary.

**2. Breach Report (40%)**

- Assess research, technical analysis, and proposed solutions.
- Includes introduction (10%), analysis (15%), and lessons learned (15%).

**3. Vocabulary Use (20%)**

- Measure correct and effective integration of at least **10 key terms** like **firewall** and **encryption**.

**4. Listening Task (10%)**

- Test ability to extract and summarize information from the video.

**Total: 100%**